



# Documenting Change



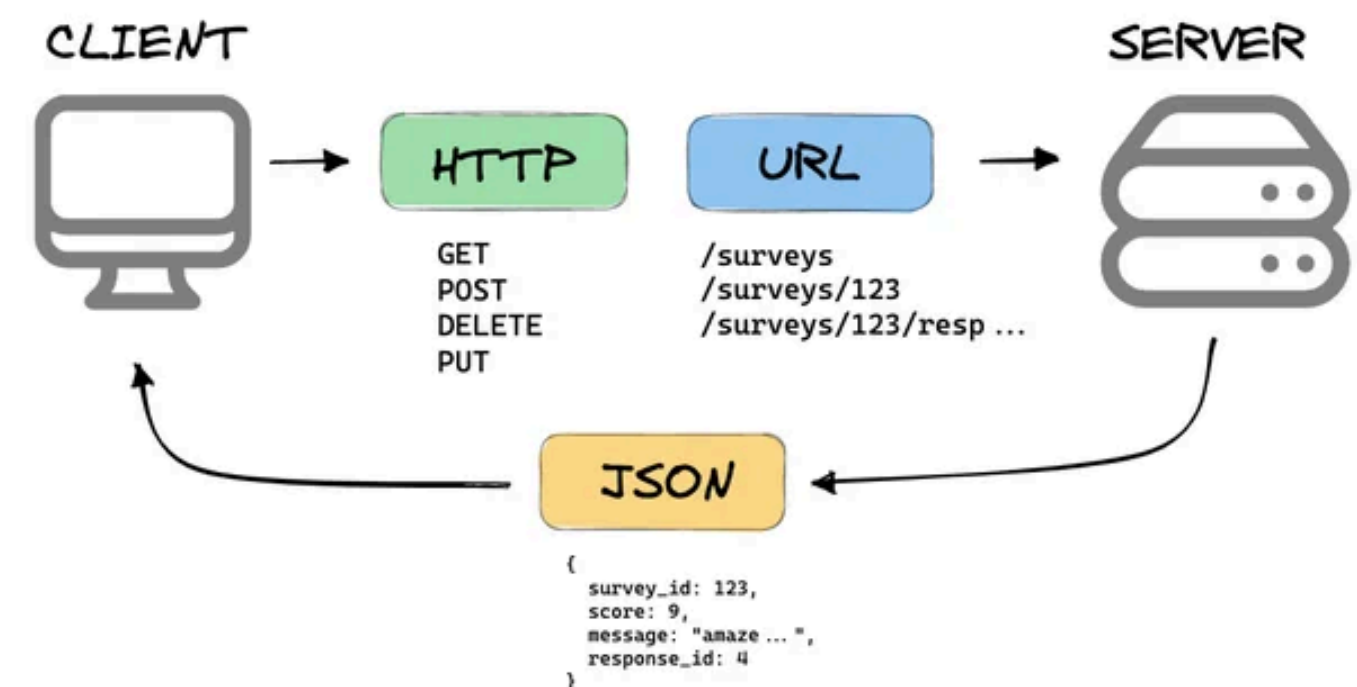


# REST API

A **REST API** (Representational State Transfer Application Programming Interface) is **an architectural style that allows different software applications to communicate with each other over the internet**. It serves as a standardized bridge between a **client** (like a mobile app or web browser) and a **server** (where data is stored), allowing them to exchange information securely and predictably.  Google Cloud +4



## WHAT IS A REST API?



mannhowie.com

**REpresentation State Transfer  
Application Program Interface**



# endpoint



a specific URL where an API can be accessed — punkt końcowy API





# request body

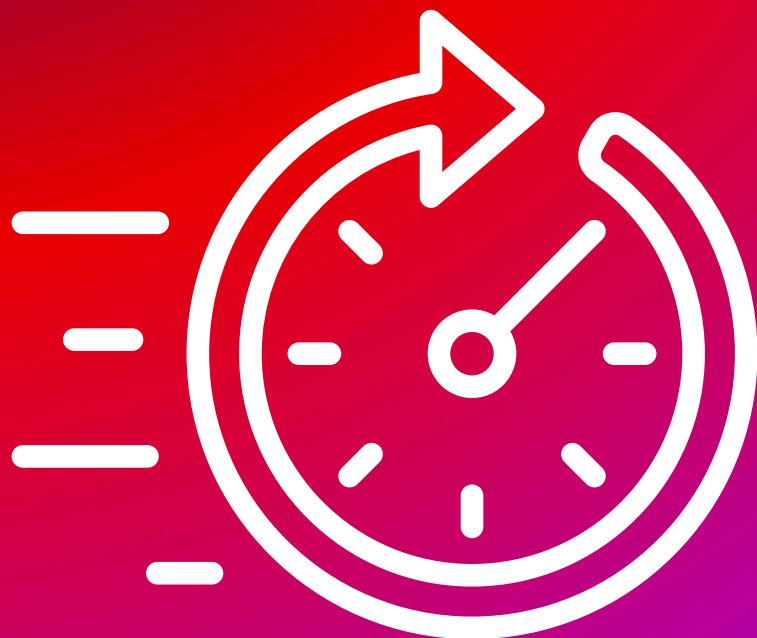


data sent by the client to the  
server in an HTTP request — treść  
żądania





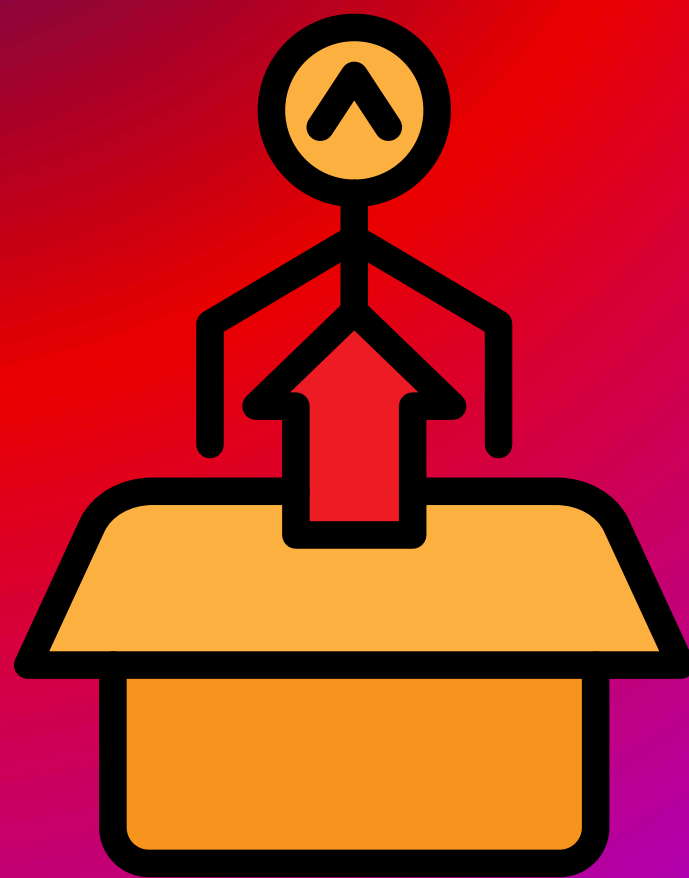
response



data returned by the server to the  
client — odpowiedź serwera



# payload

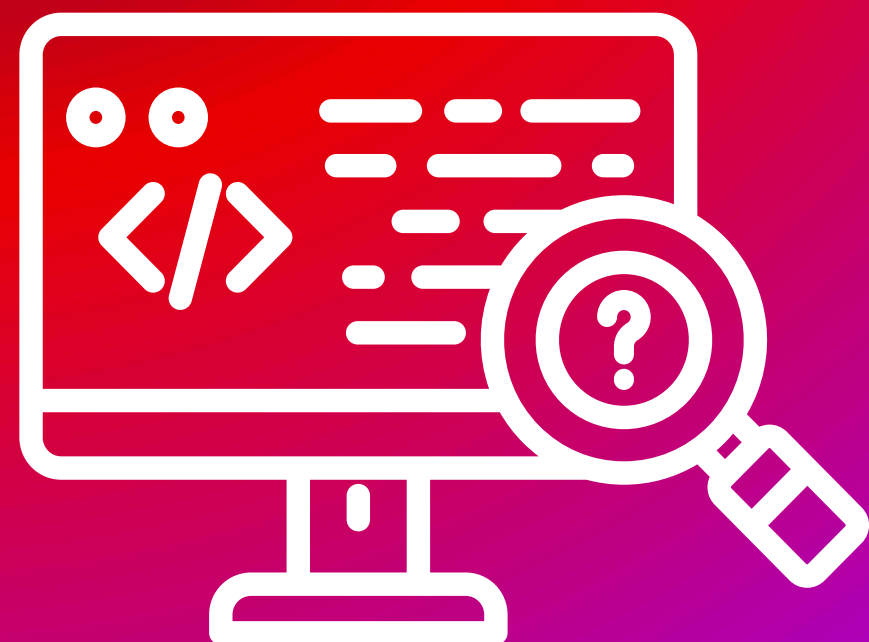


the actual data transmitted in a request or response — dane przesyłane w żądaniu / odpowiedzi





# query parameter



a value appended to a URL after "?"  
to filter or modify a request —  
parametr zapytania



# status code



a three-digit number indicating  
the result of an HTTP request — kod  
statusu HTTP





# authentication



verifying who a user is (e.g. via a token) — uwierzytelnianie

***NIGDY NIE **AUTENTYKACJA**,  
taki termin **NIE ISTNIEJE** w języku  
polskim***



## authorization



verifying what an authenticated user is allowed to do —  
autoryzacja, nadawanie uprawnień





idempotent



an operation that produces the  
same result no matter how many  
times it's repeated —  
idempotentny



# middleware



a function that processes requests  
before they reach the final handler  
— oprogramowanie pośredniczące





dependency



an external package or library  
required by the application —  
zależność (biblioteka)



deprecated

D

no longer recommended for use  
and likely to be removed in a future  
release — przestarzały,  
wycofywany





breaking  
change

**BREAKING**  
**NEWS**

a change that makes existing  
client code stop working correctly  
— zmiana łamiąca kompatybilność





## backward compatible



working correctly with previous  
versions of the software —  
kompatybilny wstecznie





## technical debt



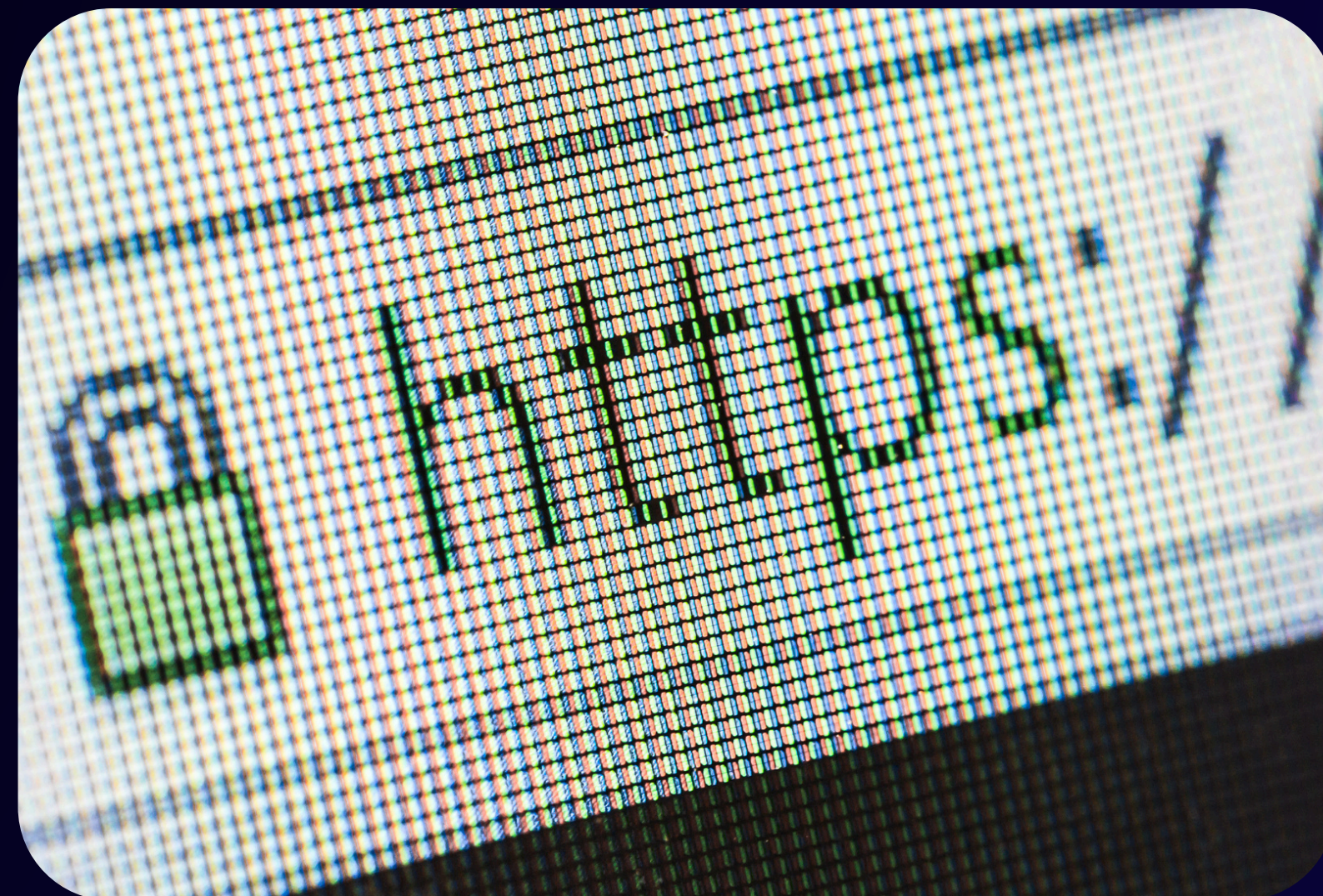
extra rework caused by choosing a quick fix over a better long-term solution — dług techniczny





# HTTP

HTTP (Hypertext Transfer Protocol) is the foundational protocol used to **transfer data across the World Wide Web**. It operates on a client-server model, allowing web browsers to request and load websites, images, and videos from remote servers.



HyperText Transfer Protocol





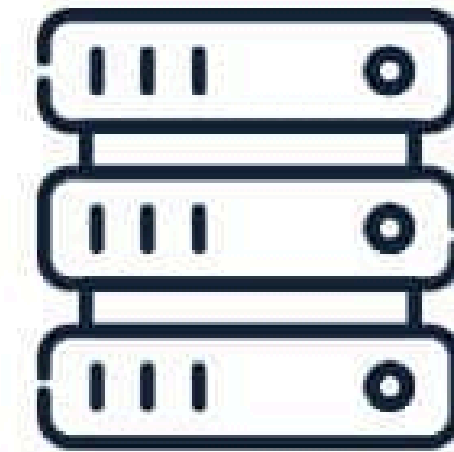
# HTTP vs HTTPS



USER



INSECURE CONNECTION



NORMAL HTTP



USER



ENCRYPTED CONNECTION



SECURE HTTPS

SSL  
CERTIFICATE



# HTTP Status Codes

2XX

SUCCESS

4XX

CLIENT  
ERROR

5XX

SERVER  
ERROR





# HTTP Status Codes

| Kod | Nazwa                 | Kategoria | Znaczenie                                                      |
|-----|-----------------------|-----------|----------------------------------------------------------------|
| 200 | OK                    | 2xx       | The request succeeded; the standard success response.          |
| 201 | Created               | 2xx       | The request succeeded and a new resource was created.          |
| 400 | Bad Request           | 4xx       | The server can't process the request (e.g. malformed JSON).    |
| 401 | Unauthorized          | 4xx       | Authentication is missing or invalid.                          |
| 403 | Forbidden             | 4xx       | Authenticated, but not allowed to perform this action.         |
| 404 | Not Found             | 4xx       | The requested resource doesn't exist.                          |
| 409 | Conflict              | 4xx       | The request conflicts with the current state (e.g. duplicate). |
| 500 | Internal Server Error | 5xx       | A generic error on the server side.                            |



# Commits

In coding and version control, a **commit** is a permanently saved snapshot of your project's files at a specific point in time. Think of it as a digital "save point" in a video game; if you make a mistake later, you can instantly roll back your codebase to this exact moment.  W3Schools +2

```
$ git commit -m "first commit"
[master (root-commit) 5a457ea] first commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file.txt
```

GIT COMMIT







# Conventional Commits

kategoria zmiany, np. feat, fix, docs (pełna lista — kolejny slajd)

scope (opcjonalny) — moduł, którego dotyczy zmiana, np. feat(auth)

«**type**»[optional **scope**]: «**description**»

[optional body]  
[optional footer(s)]

**description** — krótki opis w czasie teraźniejszym, bez kropki na końcu

**! po type/scope lub stopka BREAKING CHANGE: — oznacza zmianę łamiącą kompatybilność**



# Commit types

| Typ      | Znaczenie                                                             | Przykład                                                 |
|----------|-----------------------------------------------------------------------|----------------------------------------------------------|
| feat     | a new feature for the user (correlates with MINOR in SemVer)          | feat(auth): add password-reset endpoint                  |
| fix      | a bug fix (correlates with PATCH in SemVer)                           | fix(tasks): return 409 instead of 500 on duplicate title |
| docs     | documentation only changes                                            | docs(readme): document POST /api/tasks                   |
| style    | formatting changes that don't affect meaning (whitespace, semicolons) | style(tasks): apply prettier formatting                  |
| refactor | a code change that neither fixes a bug nor adds a feature             | refactor(tasks): extract validation into middleware      |







# Commit types

| Typ   | Znaczenie                                            | Przykład                                          |
|-------|------------------------------------------------------|---------------------------------------------------|
| perf  | a code change that improves performance              | perf(tasks): add index on due_date field          |
| test  | adding or correcting tests                           | test(tasks): add unit tests for creation endpoint |
| build | changes to the build system or external dependencies | build(deps): bump express to 4.19.2               |
| ci    | changes to CI configuration files and scripts        | ci: run test suite on pull requests               |
| chore | other changes that don't modify src or test files    | chore: update .gitignore                          |





# Coventional Commits



**Dr Pigeon**

DevOps

Change ID: 2137

**ci(docker): add container clean-up script**

This script (*cleanup.sh*) removes the container and its network before the container itself is stopped.





# Exercise – commits

| # | Oryginalny commit                                                                                | Problem                                                                                                                               |
|---|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1 | "fixed stuff"                                                                                    | No type, no scope, no indication of what was actually fixed — impossible to search or generate a changelog from this.                 |
| 2 | "added new thing for task priority, also updated readme and fixed a small bug in the tags field" | Three unrelated changes bundled into a single commit — should be split into three separate, atomic commits.                           |
| 3 | "poprawki w kontrolerze tasks"                                                                   | Written in Polish and too vague — commit messages in international projects are written in English and describe the change precisely. |
| 4 | "WIP"                                                                                            | "Work in progress" markers belong in a draft pull request title, not in the permanent commit history of the main branch.              |





# Exercise – commits

#1

`fix(tasks): correct due-date validation for past dates`

#2

`feat(tasks): add priority field to task model`  
`docs(readme): document the priority field`  
`fix(tasks): correct tags array validation`

#3

`refactor(tasks): simplify controller error handling`

#4

`feat(tasks): add initial task filtering by status`



# Changelog

A changelog is a curated, chronologically ordered file that contains a summary of notable changes made to a project during each release. While a commit history logs every single technical change, a changelog translates those technical updates into a clean, human-readable format for users, customers, or other stakeholders.



## 9.1.0 (2020-03-25)

### Release Highlights

To learn about the release highlights and our CLI-powered automated update workflow for your projects please check out the [v9.1 release announcement](#).

- TypeScript 3.8 update
- ngcc improvements
  - performance optimizations
  - concurrency & reliability improvements for monorepo use-cases (npm postinstall script no longer recommended)
- i18n now supports RTL locale info
- Ivy compatibility fixes

### Features

- **bazel:** enable ivy template type-checking in g3 ([#35672](#)) ([8f5b7f3](#))
- **bazel:** transform generated shims (in Ivy) with tsickle ([#35975](#)) ([e3ecdc6](#)), closes [#35848](#)
- **compiler-cli:** implement NgTscPlugin on top of the NgCompiler API ([#34792](#)) ([3c69442dbd](#))
- **compiler:** Add sourceSpan and keySpan to TemplateBinding ([#35897](#)) ([06779cf](#))
- **compiler:** Propagate source span and value span to Variable AST ([#36047](#)) ([31bec8c](#))
- **compiler:** add dependency info and ng-content selectors to metadata ([#35695](#)) ([fb70083](#))
- **language-service:** improve non-callable error message ([#35271](#)) ([acc483e](#))
- **language-service:** modularize error messages ([#35678](#)) ([47a1811](#)), closes [#32663](#)
- **ngcc:** implement source-map flattening ([#35132](#)) ([df816c9](#))
- **ngcc:** pause async ngcc processing if another process has the lockfile ([#35131](#)) ([eef0753](#))





# JSDoc

JSDoc is a markup language used to annotate JavaScript source code files. It allows developers to add documentation directly inside code comments, which can then be read by code editors or used to automatically generate an HTML documentation website.

```
/**
 * Registers a new pet into the Pet Adoption Center.
 *
 * @memberof Pets
 * @param {Pets.Pet} pet - The pet.
 * @returns {Promise<Transactions.TransactionResponse>} Details about the transaction.
 * @example
 * const sdk = require('pet-adoption-center');
 *
 * const transaction = sdk.pets.register({
 *   name: 'Atticus',
 *   type: sdk.PET_TYPES.DOG,
 *   breed: sdk.DOG_BREEDS.FRENCH_BULLDOG,
 *   dateOfBirth: '2010-03-10',
 *   neutered: false
 * });
 */
register: async function (pet : Pets.Pet ) {
  return {
    success: true,
    transactionId: '1234-5678-90',
    data: {
      // Placeholder for the pet object
    }
  };
},
```



# README

A README file is a text document that introduces and explains a software project, data set, or code repository. It acts as the "front door" of a project, serving as the primary guide for users and developers to understand what the project does and how to get started.



## README

Curl is a command line tool for transferring data specified with URL syntax. Find out how to use curl by reading the curl.1 man page or the MANUAL document. Find out how to install Curl by reading the INSTALL document.

libcurl is the library curl is using to do its job. It is readily available to be used by your software. Read the libcurl.3 man page to learn how!

You find answers to the most frequent questions we get in the FAQ document.

Study the COPYING file for distribution terms.

Those documents and more can be found in the docs/ directory.

## CONTACT

If you have problems, questions, ideas or suggestions, please contact us by posting to a suitable mailing list. See <https://curl.se/mail/>

All contributors to the project are listed in the THANKS document.

## WEBSITE

Visit the curl website for the latest news and downloads:

<https://curl.se/>



# Change Categories

**ADDED**

for new features

**CHANGED**

for changes in  
existing functionality

**DEPRECATED**

for soon-to-be  
removed features

**REMOVED**

for now removed  
features

**FIXED**

for any bug fixes

**SECURITY**

in case of  
vulnerabilities





# Changelog



## [1.4.0] - 2026-07-19

### Added

- Tasks can now be assigned a priority (low, normal, high).

### Fixed

- Creating a task with a duplicate title now returns 409 instead of crashing the server with a 500 error.





# JSDoc



```
/**
 * Creates a new task for the currently authenticated user.
 *
 * @param {string} req.body.title - Title of the task (must be unique per user).
 * @param {string} req.body.dueDate - ISO 8601 due date for the task.
 * @param {string} [req.body.priority='normal'] - Task priority.
 * @returns {201} Task created – returns id, title, and status.
 * @throws {409} A task with this title already exists for this user.
 */
```



# README



### POST /api/tasks

Creates a new task for the authenticated user.

**\*\*Request body:\*\***

```
{ "title": string, "dueDate": string, "priority": string }
```

**\*\*Success response:\*\*** 201 Created

```
{ "id": string, "title": string, "status": "open" }
```

**\*\*Error response:\*\*** 409 Conflict – a task with this title already exists for the current user.





APLIKACJE  
WEBOWE

:wq

L0



by **Mateusz Węglewski**